

REPORT ON FABRIC DEFECT DETECTION

(29th June 2023 - 30th June 2023)

RESEARCH PAPER:

Reviewed and comprehended the content of the research papers, a PowerPoint presentation has been created based on the findings.

	PAPER	PPT Link	Remarks
Dhruv	Fabric Defect Detection Using Computer Vision Techniques: A Comprehensive Review	 Fabric Defect Detectio...	
Jatan	Automatic Fabric Defect Detection using a deep Convolutional Neural Network	 P1_BY_JATAN	
Prachi	Fabric Defect Detection based on completed local quartet patterns and majority decision algorithm.	 Fabric Defect Detectio...	
Amit	Fabric Defect Detection based on completed local quartet patterns and majority decision algorithm.	 Review on Fabric Defe...	
Muskan	Fabric defect detection using the improved YOLOv3 model	 Fabric Defect Detectio...	

Jatan	Mobile-Unit: An efficient convolutional neural network for fabric defect detection	📄 P6_BY_JATAN	
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Image Classification:

Learned and Applied models on the MNIST Dataset of Fashion. Calculated F1 macro score for the model and tried to improve the scores.

	ML MODELS	CODE LINK	Macro- F1 Score(best)
Jatan	Image classification task on fashion MNIST dataset using TensorFlow, tuning the hyperparameter using Keras tuner and reported macro F1 score.	GitHub File	0.8801 (After tuning the parameter)
Muskan	Image classification on Mnist fashion dataset using cnn and computed precision, recall, F1 score, and support for multiple hidden layers and after tuning hyperparameters.	Colab Link	0.88
Amit	Performed Image classification task on fashion MNIST dataset using TensorFlow, tuned the hyperparameter using Keras tuner, and computed macro F1 score.	Colab Link	0.886566
Prachi	Worked on MNIST handwritten digit dataset, built the CNN model, used RandomSearch to find the best set of parameters, trained a model, and evaluated its performance.	Colab Link	0.99
Dhruv	Image Classification using CNN model, tuning hyperparameter using Keras Tuner. The F1 score is also calculated.	Colab Link	0.89

Analysis of MNIST Fashion dataset :-There is total 6 classes in this dataset

Overall F1 Score Report: ■ Report on Macro F1 Score.pdf

Overall Analysis of MNIST Fashion dataset

DATASET DESCRIPTION :-

- Fashion-MNIST is a dataset designed as a drop-in replacement for the MNIST dataset.
- The dataset contains 70,000 grayscale images of fashion-related items, each measuring 28x28 pixels.
- There are 10 different classes or labels representing different types of fashion items.
- The dataset is divided into a training set of 60,000 images and a testing set of 10,000 images, following the same structure as the original MNIST dataset..

There is total 10 classes in this dataset .

- ❖ Class 0 : T-Shirt/Top
- ❖ Class 1 : Trouser
- ❖ Class 2 : Pullover
- ❖ Class 3 : Dress
- ❖ Class 4 : Coat
- ❖ Class 5 : Sandals
- ❖ Class 6 : Shirts
- ❖ Class 7 : Sneakers
- ❖ Class 8 : Bag
- ❖ Class 9 : Ankle boots

RESULT

	precision	recall	f1-score	support
Class 0	0.82	0.84	0.83	1000
Class 1	0.98	0.97	0.98	1000
Class 2	0.87	0.69	0.77	1000
Class 3	0.87	0.91	0.89	1000
Class 4	0.71	0.85	0.78	1000
Class 5	0.97	0.98	0.98	1000
Class 6	0.68	0.65	0.66	1000
Class 7	0.94	0.97	0.95	1000
Class 8	0.99	0.96	0.98	1000
Class 9	0.98	0.94	0.96	1000
accuracy			0.88	10000
macro avg	0.88	0.88	0.88	10000
weighted avg	0.88	0.88	0.88	10000

ANALYSIS OF METRICS:

Analyzing the provided metrics, which include precision, recall, and F1-score for each class, as well as the overall accuracy, macro average, and weighted average, we can draw the following insights:

1. Precision: Classes 1, 3, 5, 7, 8, and 9 have high precision scores, indicating that the model performs well in accurately predicting these classes. Classes 0, 2, 4, and 6 have slightly lower precision scores, suggesting that the model may have more false positives for these classes.

2. Recall: Classes 1, 3, 5, 7, 8, and 9 have high recall scores, indicating that the model effectively captures most of the positive instances for these classes. Classes 0, 2, 4, and 6 have slightly lower recall scores, suggesting that the model may miss some positive instances for these classes.

3. F1-Score: Classes 1, 3, 5, 7, 8, and 9 have high F1-scores, indicating a good balance between precision and recall for these classes. Classes 0, 2, 4, and 6 have slightly lower F1-scores, suggesting a potential imbalance between precision and recall for these classes.

4. Accuracy: The overall accuracy of the model is 0.88, which indicates that the model correctly classifies approximately 88% of the instances in the test dataset.

5. Macro Average: The macro average treating each class equally. The macro average precision, recall, and F1-score are all around 0.88, which aligns with the overall accuracy.

6. Weighted Average: The weighted average precision, recall, and F1-score are also around 0.88, indicating that the model's performance is similar when considering the class distribution in the dataset.

Overall, the analysis of these metrics provides insights into the model's performance for each class in the Fashion MNIST dataset. It highlights areas of strength and areas that may require improvement. Further investigation can be done to understand the reasons behind any discrepancies in performance and to potentially fine-tune the model for better results, especially for classes with slightly lower precision, recall, and F1-scores.

DATASET DESCRIPTION

DATASET NAME	TOTAL IMAGES	IMAGE SIZE	NO. OF DEFECT CLASSES	NO. OF DEFECTS PER CLASSES	DATASET SIZE	DATASET RESOURCES
Fabric-defect-data set	412		3 DEFECTIVE 1 ERROR FREE	horizontal - 136 vertical - 92 holes - 184 ERROR FREE - 2	224 MB	https://www.kaggle.com/datasets/rmshashi/fabric-defect-dataset
Textile Defect Detection	144000	32x32 or 64x64	5 - DEFECTIVE 1 - ERROR FREE	good - 24480 DEFECTS = 119520 color thread cut hole metal_contamination	4GB	https://www.kaggle.com/datasets/belkhirnacim/textiledefectdetection
Fabric Defects	24	512x512	2 DEFECTIVE	HOLES AND CUTS - 24 IMAGES		https://github.com/msmi-nhas93/FabricDefect
Tilda dataset	3200	768 x 512	4 DIFFERENT FABRIC 7 DEFECTIVE 1 ERROR FREE	4 DIFFERENT FABRIC - 2CLASS ERROR FREE 1 CLASS (50 IMAGES PER CLASS PER FABRIC) ERROR CLASSES:- 1 - HOLE & CUTS(MECHANICAL DAMAGES) 2 - OIL & COLOUR STAIN 3 - THREAD ERROR(ABSENCE OF INDIVIDUAL THREAD)(WITHOUT MECHANICAL) 4 - FOREIGN OBJECTS (SO-CALLED FLIGHT) 5 -CREASES (WITHOUT MECHANICAL DAMAGE) 6. CHANGED LIGHTING CONDITIONS. 7.AFFINE DISTORTIONS CAUSED BY TILTING THE CAMERA AND CHANGING	1.27 GB	https://lmb.informatik.uni-freiburg.de/resources/datasets/tilda_en.html
Defect shirts	66		1 DEFECTIVE 1 ERROR FREE	DEFECTIVE - 35 ERROR FREE - 31	5.5 MB	https://universe.roboflow.com/tharshen/defect-shirts-kmeml/browse?queryText=class%3Atoarn&pageSize=50&startingIndex=0&browseQuery=true
Leather	369	1024 X 1024	5-DEFECTIVE 1 ERROR FREE	COLOR ERROR = 19, CUT ERROR = 19, FOLD ERROR = 17, GLUE ERROR = 19, POKE ERROR = 18, ERROR FREE = 277	500 MB	https://www.mvtec.com/company/research/datasets/mvtec-ad/downloads
Surface error	1150	512 X 512	1 DEFECTIVE 1 ERROR FREE	DEFECTIVE - 150 ERROR FREE - 1000	253 MB	https://hci.iwr.uni-heidelberg.de/content/weakly-supervised-learning-industrial-optical-inspection

VARIOUS CNN NETS

Several popular deep learning libraries in Python implement various convolutional neural network (CNN) architectures. Here are some commonly used CNN architectures and their corresponding Python packages:

1. LeNet-5:

- Package: Keras (with TensorFlow backend)
- It comprises seven layers, including convolutional, pooling, and fully connected layers.
- LeNet-5 was designed for handwritten digit recognition tasks and played a significant role in popularizing CNNs.

2. AlexNet:

- Package: PyTorch
- It was the winning model of the ImageNet Large-Scale Visual Recognition Challenge (ILSVRC) in 2012.
- AlexNet features a deep network with multiple convolutional and fully connected layers, utilizing techniques such as ReLU activation and dropout regularization.

3. VGGNet:

- Package: Keras (with TensorFlow backend)
- It consists of multiple convolutional layers with small 3x3 filters, followed by max-pooling layers.
- VGGNet achieved outstanding performance in various image classification tasks and served as a baseline for subsequent architectures.

4. ResNet:

- Package: PyTorch
- It addresses the problem of vanishing gradients in very deep networks.
- It utilizes residual connections to enable the network to learn residual mappings.
- ResNet architectures, such as ResNet-50 and ResNet-101, have achieved state-of-the-art results in image recognition tasks.

5. GoogLeNet (Inception):

- Package: Keras (with TensorFlow backend)
- It utilizes multiple parallel convolutional layers of different sizes to capture different levels of information.
- GoogLeNet is known for its computational efficiency and achieved high accuracy in the ILSVRC 2014 competition.

6. DenseNet:

- Package: PyTorch
- is based on the idea of densely connected layers.
- It connects each layer to every other layer in a feed-forward fashion, promoting feature reuse and enhancing gradient flow.
- DenseNet architectures are compact and have achieved competitive performance while reducing the number of parameters.

7. MobileNet:

- Package: Keras (with TensorFlow backend)
- MobileNet is a family of CNN architectures designed for mobile and resource-constrained devices.
- It employs depthwise separable convolutions to reduce computation and model size while maintaining accuracy.
- MobileNet models strike a good balance between efficiency and performance.

8. SqueezeNet:

- Package: PyTorch
- SqueezeNet is another lightweight CNN architecture designed for resource-constrained environments.
- It utilizes 1x1 pointwise convolutions to reduce model size and employs bypass connections to retain high-level features.
- SqueezeNet achieves comparable accuracy to larger models with a significantly smaller model size.

9. ResNet (Residual Network):

- Package: PyTorch
- It introduces residual connections.
- By learning residual mappings, ResNet enables the training of deep networks with improved optimization and accuracy.
- ResNet has become a fundamental architecture in deep learning and has been widely adopted in various domains.

10. ResNeXt:

- Package: PyTorch
- ResNeXt is an extension of ResNet that introduces the concept of a "cardinality" parameter.
- It utilizes grouped convolutions to split the input channels into multiple paths, enabling the model to capture diverse feature representations.
- ResNeXt has demonstrated improved accuracy compared to ResNet on various visual recognition tasks.

11. EfficientNet:

- Package: Keras (with TensorFlow backend)
- It employs a compound scaling method to balance model depth, width, and resolution, resulting in efficient and accurate models.
- EfficientNet models have achieved state-of-the-art performance on various image classification and detection tasks.

Nets that are mainly used in Textile Industries :

- MobileNet
- EfficientNet
- VGG (Visual Geometry Group)
- ResNet (Residual Network)
- InceptionNet (GoogLeNet)